harriet-mason / ARSA-UncertaintyLitReview
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings

Comparing changes

This is a direct comparison between two commits made in this repository or its related repositories. View the default comparison for this range or learn more about diff comparisons.

base: 268d598
compare: 624b544

Showing 18 changed files with 2,876 additions and 775 deletions.

Split
Unified

```
4 .gitignore
BIN +328 KB Comparing 268d598..17c70bd · harriet-mason:ARSA-UncertaintyLitReview.pdf
Binary file not shown.
BIN +1 MB ST14-D-Cook-Orig-Reviewed-Reply.pdf
3 _extensions/christopherkenny/annual-reviews/partials/before-bib.tex
1,046 ar-1col-S20.cls
1 ar-style1.bst
859 changes.diff
7 compare-commits.txt
35 distcode.R
BIN +256 KB (150%) paper.pdf
Binary file not shown.
492 paper.qmd
@@ -52,16 +52,13 @@ author:
        department: Department of Mathematics & Statistics
        city:  Logan, UT
        country: United States
- abstract: Better representation of the uncertainty in a data visualisation is a focus of recent research activity. A problem with the current literature is that there is a lack of clarity about the definition of uncertainty and what it means to represent it in a plot. This confusion results in a significant amount of conflicting results in the literature, especially in experiments that assess the effectiveness of different uncertainty representations. In this review, we summarise the current literature, provide workable definitions, and illustrate these definitions with examples. In doing so, we ask what it really takes to achieve transparency in statistical graphics. It is hoped that it will be useful for guiding new graphics methodology and experimental research.
+
+ abstract: Better representation of the uncertainty in a data visualisation is a focus of recent research activity. A problem with the current literature is that there is a lack of clarity about the definition of uncertainty and what it means to represent it in a plot. This confusion has led to conflicting results in the literature, especially in experiments that assess the effectiveness of different uncertainty representations. In this review, we summarise the current literature, provide workable definitions, and illustrate these definitions with examples. In doing so, we ask what it really takes to achieve transparency in statistical graphics. It is hoped that it will be useful for guiding new graphics methodology and experimental research.
  date: last-modified
- toc: false
- number-sections: true
- latex-clean: true
  format:
-   jasa-pdf:
+   annual-reviews-pdf:
      keep-tex: true
-     journal:
-       blinded: false
+     toc: false
      filters:
      - at: pre-quarto
        path: _extensions/andrewheiss/wordcount/citeproc.lua
@@ -77,7 +74,8 @@ format:
  fig-valign: bottom
  cap-location: bottom
  bibliography: paper.bib
- keywords: [data visualisation, statistical graphics, data science, information visualisation]
+ suppress-bibliography: true
+ keywords: [data visualisation, statistical graphics, data science]
  journal-name: Annual Review of Statistics and Its Application
  journal-volume: AA
  journal-year: 2023
@@ -214,7 +212,7 @@ reg1a <- mtcars |>
                colour = "dodgerblue3",
                linewidth = 1.5) +
    theme(aspect.ratio=1) +
-   ggtitle("R1 - low") +
+   ggtitle("R1 - low uncertainty") +
    ylim(9,35) +
    xlim(1.5,5.5)

@@ -226,7 +224,7 @@ reg1b <- small_cars |>
                slope = slope_est_2,
                colour = "dodgerblue3",
                linewidth = 1.5) +
-   ggtitle("R2 - high") +
+   ggtitle("R2 - high uncertainty") +
    theme(aspect.ratio=1) +
    ylim(9,35) +
    xlim(1.5,5.5)
@@ -239,7 +237,7 @@ reg2a <- augment(fit1) |>
    theme_few()  +
    #geom_smooth(linewidth = 1.5,
    #            colour = "dodgerblue3", se=FALSE) +
-   ggtitle("R1 - low") +
+   ggtitle("R1 - low uncertainty") +
    ylab("residual") +
    theme(aspect.ratio=1) +
    ylim(-5.5,7) +
@@ -252,7 +250,7 @@ reg2b <- augment(fit2) |>
    theme_few()  +
    #geom_smooth(linewidth = 1.5,
    #            colour = "dodgerblue3", se=FALSE) +
-   ggtitle("R2 - high") +
+   ggtitle("R2 - high uncertainty") +
    ylab("residual") +
    theme(aspect.ratio=1) +
    ylim(-5.5,7) +
@@ -269,7 +267,7 @@ reg3a <- mtcars |>
                alpha = 2/int_se_1,
                linewidth = 2*slope_se_1,
                colour = "dodgerblue3") +
-   ggtitle("R1 - low") +
+   ggtitle("R1 - low uncertainty") +
    theme(aspect.ratio=1) +
    ylim(9,35) +
    xlim(1.5,5.5)
@@ -283,7 +281,7 @@ reg3b <- small_cars |>
                alpha = 2/int_se_2,
                linewidth = 2*slope_se_2,
                colour = "dodgerblue3") +
-   ggtitle("R2 - high") +
+   ggtitle("R2 - high uncertainty") +
    theme(aspect.ratio=1) +
    ylim(9,35) +
    xlim(1.5,5.5)
@@ -298,7 +296,7 @@ reg4a <-  mtcars |>
                       slope = dist_normal(slope_est_1,slope_se_1),
                       alpha=0.2, times=100,
                       colour = "dodgerblue3")+
-   ggtitle("R1 - low") +
+   ggtitle("R1 - low uncertainty") +
    theme(aspect.ratio=1) +
    ylim(9,35) +
    xlim(1.5,5.5)
@@ -311,7 +309,7 @@ reg4b <- small_cars |>
                       slope = dist_normal(slope_est_2, slope_se_2),
                       alpha=0.5, times=100,
                       colour = "dodgerblue3")+
-   ggtitle("R2 - high") +
+   ggtitle("R2 - high uncertainty") +
    theme(aspect.ratio=1) +
    ylim(9,35) +
    xlim(1.5,5.5)
@@ -337,7 +335,7 @@ uni1a <- group_df |>
    ggplot() +
    geom_point(aes(x=mean(dist_low), y=group), size = 3, colour = "forestgreen") +
    theme_few() +
-   ggtitle("G1 - low") +
+   ggtitle("G1 - low uncertainty") +
    labs(y = "treatment group",
         x = "treatment effect") +
    theme(aspect.ratio=1)
@@ -346,7 +344,7 @@ uni1b <- group_df |>
    ggplot() +
    geom_point(aes(x=mean(dist_high), y=group), size = 3, colour = "forestgreen") +
    theme_few() +
-   ggtitle("G2 - high") +
+   ggtitle("G2 - high uncertainty") +
    labs(y = "treatment group",
         x = "treatment effect") +
    theme(aspect.ratio=1)
@@ -356,7 +354,7 @@ uni2a <- group_df |>
    ggplot() +
    geom_point(aes(x=round(variance(dist_low), 4), y=group), size = 3, colour = "forestgreen") +
    theme_few() +
-   ggtitle("G1 - low") +
+   ggtitle("G1 - low uncertainty") +
    labs(y = "treatment group",
         x = "variance of treatment effect") +
    theme(aspect.ratio=1)
@@ -365,7 +363,7 @@ uni2b <- group_df |>
    ggplot() +
    geom_point(aes(x=variance(dist_high), y=group), size = 3, colour = "forestgreen") +
    theme_few() +
-   ggtitle("G2 - high") +
+   ggtitle("G2 - high uncertainty") +
    labs(y = "treatment group",
         x = "variance of treatment effect") +
    theme(aspect.ratio=1)
@@ -375,7 +373,7 @@ uni3a <- group_df |>
    ggplot(aes(xdist = dist_low, y=group)) +
    theme_few() +
    stat_pointinterval(.width = c(.95), colour = "forestgreen") +
-   ggtitle("G1 - low") +
+   ggtitle("G1 - low uncertainty") +
    labs(y = "treatment group",
         x = "treatment effect") +
    theme(aspect.ratio=1)
@@ -384,7 +382,7 @@ uni3b <- group_df |>
    ggplot(aes(xdist = dist_high, y=group)) +
    theme_few() +
    stat_pointinterval(.width = c(.95), colour = "forestgreen") +
-   ggtitle("G2 - high") +
+   ggtitle("G2 - high uncertainty") +
    labs(y = "treatment group",
         x = "treatment effect") +
    theme(aspect.ratio=1)
@@ -394,7 +392,7 @@ uni4a <- group_df |>
    ggplot(aes(xdist = dist_low, y=group)) +
    theme_few() +
    stat_dots(fill = "forestgreen", colour=NA) +
-   ggtitle("G1 - low") +
+   ggtitle("G1 - low uncertainty") +
    labs(y = "treatment group",
         x = "treatment effect") +
    theme(aspect.ratio=1)
@@ -403,7 +401,7 @@ uni4b <- group_df |>
    ggplot(aes(xdist = dist_high, y=group)) +
    theme_few() +
    stat_dots(fill = "forestgreen", colour=NA) +
-   ggtitle("G2 - high") +
+   ggtitle("G2 - high uncertainty") +
    labs(y = "treatment group",
         x = "treatment effect") +
    theme(aspect.ratio=1)
@@ -430,10 +428,10 @@ choro1 <- my_map_data |>
          legend.direction = "horizontal",
         legend.background = element_rect(fill=NA),
         plot.title = element_text(size = 14)) +
-   ggtitle("S1 - low")
+   ggtitle("S1 - low uncertainty")

  choro2 <- choro1 %+% filter(my_map_data, variance_class=="highvar") +
-   ggtitle("S2 - high")
+   ggtitle("S2 - high uncertainty")

  #####################  Exceed Prob Map #####################

@@ -455,10 +453,10 @@ exceed1 <- exeed_data |>
         legend.direction = "horizontal",
         legend.background = element_rect(fill=NA),
         plot.title = element_text(size = 14)) +
-   ggtitle("S1 - low")
+   ggtitle("S1 - low uncertainty")

  exceed2 <- exceed1 %+% filter(exeed_data, variance_class=="highvar") +
-   ggtitle("S2 - high")
+   ggtitle("S2 - high uncertainty")

  ##################### Bivariate maps #####################

@@ -486,10 +484,10 @@ bivar1 <- my_map_data |>
    scale_fill_manual(values = bivariatepal) +
    theme_void() +
    theme(legend.position = "none") +
-   ggtitle("S1 - low")
+   ggtitle("S1 - low uncertainty")

  bivar2 <- bivar1 %+% filter(my_map_data, variance_class=="highvar") +
-   ggtitle("S2 - high")
+   ggtitle("S2 - high uncertainty")


  bivar_pal<- expand_grid(bitemp = seq(1,8), bivar = seq(1,4)) |>
@@ -513,8 +511,11 @@ bivar3 <- bivar_pal |>

  #bivar1 <- bivar1 / bivar3
  #bivar2 <- bivar2 / bivar3
- bivar1 <- bivar1 + bivar3 + plot_layout(ncol=1, heights = c(4, 1))
- bivar2 <- bivar2 + bivar3 + plot_layout(ncol=1, heights = c(4, 1))
+ bivar1 <-  bivar1 + inset_element(bivar3, left = 0.55, bottom = -0.5,
+                           right = 1, top = 0.2)
+
+ bivar2 <-  bivar2 + inset_element(bivar3, left = 0.55, bottom = -0.5,
+                           right = 1, top = 0.2)

  ##################### VSUP maps #####################
  # Function to combine colours for VSUP
@@ -585,8 +586,11 @@ vsup4 <- vsup3 +

  #vsup1 <- vsup1 / vsup3
  #vsup2 <- vsup2 / vsup4
- vsup1 <- vsup1 + vsup3 + plot_layout(ncol=1, heights = c(4, 1))
- vsup2 <- vsup2 + vsup4 + plot_layout(ncol=1, heights = c(4, 1))
+ vsup1 <- vsup1 + inset_element(vsup3, left = 0.55, bottom = -0.5,
+                           right = 1, top = 0.2)
+
+ vsup2 <- vsup2 + inset_element(vsup3, left = 0.55, bottom = -0.5,
+                           right = 1, top = 0.2)

  ##################### Pixel maps #####################
  # low variance
@@ -604,7 +608,7 @@ pixel1 <- my_map_data |>
         legend.direction = "horizontal",
         legend.background = element_rect(fill=NA),
         plot.title = element_text(size = 14)) +
-   ggtitle("S1 - low")
+   ggtitle("S1 - low uncertainty")

  # high variance
  pixel2 <- my_map_data |>
@@ -621,27 +625,27 @@ pixel2 <- my_map_data |>
         legend.direction = "horizontal",
         legend.background = element_rect(fill=NA),
         plot.title = element_text(size = 14)) +
-   ggtitle("S2 - high")
+   ggtitle("S2 - high uncertainty")
  ```

- ## Introduction
+ # Introduction
  What do we mean when we talk about "uncertainty visualisation"?
- The phrase can feel contradictory to anyone familiar with the term.
+ The phrase can feel contradictory to anyone familiar with the individual terms.
  Among statisticians, "uncertainty" is often discussed as an omnipresent spectre touching every stage of our analysis without ever being fully seen.
  Authors will often mention that the phrase is vague [@Spiegelhalter2017; @Griethe2006], or avoid defining it by describing a list of things uncertainty *could* be [@Kinkeldey2014; @Hullman2016], but rarely do authors attempt to discuss what uncertainty actually *is*.
- By contrast, visual statistics (information visualisations, data plots) are one of the most powerful tools in the statistician's toolbox, allowing for quick and memorable communication that identifies quirks in our data that we didn't even know to look for.
+ By contrast, visual statistics (which includes information visualisations and data plots) are one of the most powerful tools in the statistician's toolbox, allowing for quick and memorable communication that identifies quirks in our data that we didn't even know to look for.
  We see this in datasets such as Anscombe's quartet [@anscombe] or the Datasaurus Dozen [@datasaurus; @datasaurpkg], where visual statistics are able to highlight elements of the data that are invisible to the typical summary statistics.
  We also see this in recall experiments, where simply sketching a distribution before recalling statistics or making predictions can greatly increase the accuracy of those measures [@Hullman2018; @Goldstein2014].
- Taken together, uncertainty visualisation implies a need to pull back the curtain and explore the unknowns of our analysis.
+ Visualisation of uncertainty enables us to pull back the curtain and explore the unknowns of our analysis.

  As nice as this sentiment is, it turns out to be easier said than done.
- Reviews on uncertainty visualisation rarely offer tried and tested rules for effective uncertainty visualisation, instead commenting on the difficulties faced when trying to summarise the field.
- @Kinkeldey2014 found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisations, or a greater goal of describing general principles.
- @Hullman2016 noticed there is a serious noise issue in the field, with noise from participants misunderstanding visualisations, misinterpreting questions, and incorrectly applying heuristics, overwhelming any information we can glean from studies.
- @MacEachren2005 identified so much contradicting evidence that they spent an entire page discussing the conflicting evidence for the question "Should I map uncertainty to colour hue?"
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+ @Kinkeldey2014 found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisation, or description of general principles.
+ @Hullman2016 noticed there is a serious noise issue in the field, with errors from participants misunderstanding visualisations, misinterpreting questions, and incorrectly applying heuristics, overwhelming any information we can glean from studies.
+ @MacEachren2005 identified so many contradictions that they spent an entire page discussing the conflicting evidence for the question "Should I map uncertainty to colour hue?".
  @Spiegelhalter2017 concluded that different plots are good for different things, arguing against a universal best plot for all people and circumstances.
- @uncertchap2022 summarised several cognitive effects that repeatedly arise in uncertainty visualisation experiments; these effects were each discussed in isolation as a list of considerations rather than an overarching theory for effective uncertainty visualisation.
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  > "Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house." - Henri Poincaré (1905)

@@ -652,85 +656,89 @@ The field is in desperate need of a unifying theory that can tie the conflicting
  This review attempts to address this issue by offering a novel perspective on the uncertainty visualisation problem.
  That is, we will use the wealth of established stone to construct a foundation to build a house.

- ## The purpose of uncertainty visualisation
+ # The purpose of uncertainty visualisation
  Mentions of "uncertainty visualisation" start springing up around 1990, across several different fields [@Ibrekk1987; @MacEachren1992], each with its own motivation for the work.
  In computer science, the area appears to be motivated by issues in the public's perception of random variables, with the hope that visualisations would give laypeople the ability to extract important information from graphical representations [@Ibrekk1987].
  With similar concerns about the public's understanding of randomness, the fields of psychology, statistics, and economics used "uncertainty visualisations" as a communication tool to mitigate the psychological bias associated with the communication of risk, a topic of concern since the early 1980s [@Spiegelhalter2017].
  In cartography, it was motivated by the inherent uncertainty of geoscience data, the practical use of visualisation as an exploratory tool, and the constrained visual channels from map representations [@MacEachren1992].
  These disparate motivations have blended together, and today, uncertainty visualisation is usually motivated by the vague goal of "decision-making".
  This term has been used to mean the mitigation of psychological bias to ensure economically rational decisions [@Padilla2022; @Padilla2021; @Kale2021], the facilitation of trust or confidence [@Zhao2023; @Yang2023], the ability to extract values related to a distribution [@Sarma2023], the prevention of false discovery in plots [@Sarma2024; @Koonchanok2023], or the extraction of some other metric that is vaguely related to "uncertainty" [@Chakraborty2024; @Ndlovu2023].
- This gradual scope creep of the field, motivated by the hazy definition of "decision-making", is the most likely culprit for the jumbled literature that makes up the field today.

- Given that there is so much subliminal disagreement in uncertainty visualisation, how did these disparate motivations come to be seen as interchangeable?
+ Given that there is so much disagreement in uncertainty visualisation, how did these disparate motivations come to be seen as interchangeable?
  All discussions on uncertainty visualisation seem to have a common thread that connects them: the belief that the ultimate goal of uncertainty visualisation is not trust, rationality, or value extraction, but *transparency*.
  We see it said directly in reviews of the field [@uncertchap2022], or when authors claim that failing to include uncertainty is akin to fraud or lying [@Hullman2020a; @Manski2020].
  We see it when authors assert that uncertainty communicates the legitimacy (or illegitimacy) of the conclusion drawn from visual inference [@Kale2018; @Griethe2006].
  We see it when authors say uncertainty visualisations should communicate a degree of confidence [@Correll2018; @Boukhelifa2012] or validity [@Hullman2020a; @Griethe2006] in our conclusions.
- We see it when authors suggest uncertainty visualisation should "guide, qualify, or soften our judgements of uncertain data" (e.g. @Leland2005 in his seminal work on the grammar of graphics).
+ We see it when authors suggest uncertainty visualisation should "guide, qualify, or soften our judgements of uncertain data" (e.g., @Leland2005 in his seminal work on the grammar of graphics).
  These authors are not wrong about the need for transparency in science communication: a six-month survey of anti-mask groups on Facebook during the COVID-19 pandemic showed that anti-maskers made persuasive arguments by exploiting inherent uncertainty ignored by pro-maskers [@Lee2021].
+ Therefore, if transparency is necessary and important, what does it look like?

  Uncertainty visualisation is motivated by the need for a sort of “visual hypothesis test”, a sentiment expressed by some authors directly [@Correll2014; @MacEachren1992].
  A successful uncertainty visualisation would act as a “statistical hedge” for any inference we make using the graphic.
  Since the purpose of a visualisation is to give a quick gist of the information [@Spiegelhalter2017], this hedging should be communicated visually without the need for complicated mental calculations.
  Therefore, an effective uncertainty visualisation should not just "show" uncertainty; untrustworthy conclusions should *not be visible*.
- If we refer to the conclusion we draw from a graphic as its signal, where uncertainty should make this signal harder to read as the "noise" increases, we can summarise the above information into three key requirements.
- A good uncertainty visualisation should:
+ If we refer to the conclusion we draw from a graphic as its "signal", and the uncertainty that should make this signal harder to read as the "noise", we can summarise the above information into three key requirements.
+ A good uncertainty visualisation should perform:

-   1) Reinforce justified signals to encourage confidence in results.
-   2) Hide spurious signals that are overwhelmed by noise.
-   3) Perform tasks 1) and 2) in a way that is proportional to the level of confidence in those conclusions.
-
- Usually, visualisations that are unconcerned with uncertainty have no issue showing justified signals, but struggle with the display of unjustified signals.
- Therefore, we **suggest calling this approach to uncertainty visualisation "signal-suppression"** since it primarily differentiates itself from the normal "noiseless" visualisation approach through criterion (2).
+   1) Signal enhancement: reinforcing statistically significant signals to encourage confidence in results, and
+   2) Signal suppression: hiding spurious signals that are overwhelmed by noise.
+
+ Additionally, the performance of tasks 1) and 2) should be proportional to the level of confidence in those conclusions.
+ When correctly performing uncertainty visualisation, we should feel reassured that what we see is actually there.
+ Therefore, we **suggest calling this approach to uncertainty visualisation "signal modulation"**.
  This is the main criterion we will use to assess the current literature on uncertainty visualisation.

- ## Current Approaches
- ### Ignoring uncertainty
+ # Current Approaches
+ ## Ignoring uncertainty
  The most common way to visualise uncertainty is to simply not.
  A study conducted by @Hullman2020a found that only a quarter of authors surveyed included uncertainty in 50% or more of their visualisations, in part because authors are not sure how to calculate uncertainty.
- This is not entirely unreasonable, given that even visualisation authors themselves seem to be in conflict about what exactly uncertainty is.
+ This is not entirely unreasonable, given that even uncertainty visualisation researchers themselves seem to be in conflict about what exactly uncertainty is.
  We will start with visualisations that ignore uncertainty with the hope that by looking at where uncertainty isn't, we can better understand where it is.

- #### What is uncertainty?
+ ### What is uncertainty?
  It is surprisingly hard to describe what uncertainty is.
  Most authors avoid the problem and describe the many characteristics of uncertainty.
- Often, uncertainty is split by factors such as whether it is due to true randomness or a lack of knowledge [@Begg2014; @Spiegelhalter2017; @Gustafson2019; @uncertchap2022; @Hullman2016; @utypo]; quantifiable or unquantifiable [@Spiegelhalter2017; @utypo; @uncertchap2022]; scientific or human [@Benjamin2018; @Gustafson2019]; systematic or random [@Sanyal2009]; statistical or bounded [@Gschwandtnei2016; @Olston2002]; accuracy or precision [@Griethe2006; @Benjamin2018; @Hullman2016]; etc.
- There are enough qualitative descriptors of uncertainty to fill a
- paper, but none of this is particularly helpful in understanding how to integrate it into a
+ Often, uncertainty is split by factors such as whether it is due to true randomness or a lack of knowledge [@Begg2014; @Spiegelhalter2017; @Gustafson2019; @uncertchap2022; @Hullman2016; @utypo]; quantifiable or unquantifiable [@Spiegelhalter2017; @utypo; @uncertchap2022]; scientific or human [@Benjamin2018; @Gustafson2019]; systematic or random [@Sanyal2009]; statistical or bounded [@Gschwandtnei2016; @Olston2002]; due to a lack of accuracy or precision [@Griethe2006; @Benjamin2018; @Hullman2016]; etc.
+ There are enough qualitative descriptors of uncertainty to fill a paper, but none of this is particularly helpful in understanding how to integrate it into a
  visualisation.

  Rather than trying to define uncertainty by looking at the myriad ways in which it *does* appear in an analysis, we may find it easier to look at where it *does not*.
  Descriptive statistics describe our sample as it is and summarise large data into a usable format, but they are not seen as the primary goal of modern statistics.
- In 19th-century England, *positivism* was the popular philosophical approach to science (positivists included famous statisticians such as Francis Galton and Karl Pearson).
+ In 19th-century England, *positivism* was the popular philosophical approach to science.
  Practitioners of the approach believed statistics ended with descriptive statistics, as science must be based on actual experience and observations [@Otsuka2023].
+ Positivists included famous statisticians such as Francis Galton and Karl Pearson.
  In order to make statements about population statistics, future values, or new observations, we need to perform inference, which requires the assumption of the "uniformity of nature", that is, we need to assume that unobserved phenomena should be similar to observed phenomena [@Otsuka2023].
- Positivists believed referencing the unobservable was bad science, embracing descriptive statistics due to the inherent certainty associated with them.
- Since uncertainty is nonexistent in descriptive statistics, it is clear that uncertainty is a by-product of inference: uncertainty is the noise that is both inseparable from our inference and meaningless without it.
-
- Rather than extracting just one element of the distribution, if you can retain the whole distribution, that not only allows the uncertainty calculation to be reproduced, but also makes it possible to derive other estimates as well.
- I think it's also important to acknowledge that if you use this approach, you need to know the "chicken" (i.e. distribution) that gave birth to the "egg" (uncertainty estimate). In practice, uncertainties are sometimes calculated by other people or organisations, and the process used to derive them may not be known.
-
-
+ Positivists believed that referencing the unobservable was bad science and that one should not make claims beyond descriptive statistics.
+ Since uncertainty is non-existent in descriptive statistics, it is clear that uncertainty is a by-product of inference: uncertainty is the noise that is both inseparable from our inference and meaningless without it.
  If we consider uncertainty to be a by-product of statistical inference, then uncertainty visualisations are the plots that depict an estimate, and therefore have an associated uncertainty.
  The most complete description of these estimates is their distributions.
+
  Rather than extracting just one element of the distribution, if you can retain the whole distribution, that not only allows the uncertainty calculation to be reproduced, but also makes it possible to derive other estimates as well.
  Suggesting distributions as a representation of uncertainty is not new.
- @Kay2023 originally suggested thinking about uncertainty visualisations as visualisations with distribution inputs, to replace the commonly used mean and standard deviation, removing the assumption of a Gaussian distribution.
- In practice, uncertainties are sometimes calculated by other people or organisations, and the process used to derive them may not be known.
+ @Kay2023 originally suggested thinking about uncertainty visualisations as visualisations with distribution inputs, to replace the commonly used mean and standard deviation and remove the assumption of a Gaussian distribution.
+ In practice, uncertainties are sometimes calculated by other people or organisations, and the process used to derive them may not be known.
  While some researchers believe these abstract notions of uncertainty, such as credibility [@Thomson2005], forecaster confidence [@Padilla2021], or uncertainty about uncertainty [@Hadjimichael2024], are too complex to be quantified, this is not necessarily true.
  Abstract concepts such as human belief or credibility are regularly quantified by Bayesians, and hierarchical approaches are often used to model uncertainty about uncertainty.

- #### Example: ignoring uncertainty
+ ### Example: ignoring uncertainty

  If visualising uncertainty is fundamentally visualising a set of random variables, what does "ignoring" uncertainty look like?
- @fig-ignore shows plots of data from three different scenarios, each with either high or low uncertainty. The plots show the expected value of the input distribution.
- In plots R1 and R2, the expected value is the line from a simple linear regression on a car's miles per gallon (`mpg`) and weight (`wt`) using the `mtcars` data, or a subset of it (available in `ggplot2` [@ggplot2], and originally from @mtcars), with differing sample sizes creating differing levels of uncertainty.
+ @fig-ignore shows plots of data from three different scenarios, each with either high or low uncertainty.
+ The plots show the expected value of the input distribution.
+ In plots R1 and R2, the expected value is the point estimate from a simple linear regression on a car's miles per gallon (`mpg`) and weight (`wt`) using the `mtcars` data (available in `ggplot2` [@ggplot2], and originally from @mtcars).
+ For the low uncertainty case, the linear regression is calculated on the full `mtcars` data, but in the high uncertainty case, it is calculated using a subset of three points.
+ The points used to calculate the linear regression are also shown on the plot; however, the fitted line is the only component that should include any uncertainty.
  Plots S1 and S2 show a choropleth map of Iowa, where counties are coloured according to a simulated temperature measurement, with measurement error being the uncertainty associated with the measuring instrument.
- Plots G1 and G2 show samples simulated from five populations (A-E). In G1, distributions have the same low variance, and in G2, they have the same high variance.
- In the linear regression, we see a downward trend for both levels of uncertainty. Here, we elected to plot the data under the fit, to provide context for this example, and help follow the thinking through to the later illustrations. Often, only the fitted line is shown. In the map, you should see a sine wave spatial trend, and in the univariate distributions, an incremental increase in treatment.
- If we were to ask a reader, "Can you see a difference between the plots in the top row versus the bottom row?
- Is the strength of the trend communicated through the visualisation?"
- The answer to both of these questions will be no for S1, S2, G1, and G2, as the high and low variance cases are identical. For R1 and R2, the answer would likely only relate to the points in the plots, not the regression line.
+ In S1 the measurement error is low, while in S2 the measurement error is high.
+ Plots G1 and G2 show samples simulated from five populations (A-E).
+ In G1, the distributions have a low variance, and in G2, they have a high variance.
+ In the linear regression, we see a downward trend, in the map, we see a sine wave, and in the univariate distributions, an incremental increase in treatment effect.
+ The noise in the high uncertainty case is set such that it should overwhelm the trend in all three
```
```

~~~diff
plots.
+ Can you see a difference between the plots in the top row versus the bottom row?
+ Is the strength of the trend communicated through the visualisation?
+ The answer to both of these questions, for S1, S2, G1, and G2, is no, as the high and low variance cases are identical.
+ For R1 and R2, any ability to differentiate the plots would likely relate to the data points, rather than the regression line.
+ The lack of differentiability between the high and low uncertainty cases highlights the danger of ignoring uncertainty in our visualisations.

  ```{r}
  #| echo: false
@@ -741,27 +749,39 @@ The answer to both of these questions will be no for S1, S2, G1, and G2, as the
  #| fig-width: 10
  #| out-width: 100%
  #| fig-pos: t
- #| fig-cap: "Three example types of data and associated plots that will be used to illustrate various choices of uncertainty representation throughout the paper: scatterplot and regression line (R1, R2), spatial choropleth map (S1, S2), grouped dotplot (G1, G2). Two levels of uncertainty (low, high) are used with each example. Ignoring the points in R1 and R2, there are no differences between the high and low uncertainty versions. Ignoring uncertainty can lead to misrepresentation of data."
+ #| fig-cap: "Three example types of data and associated plots that will be used to illustrate various choices of uncertainty representation throughout the paper: scatterplot and regression line (R1, R2), spatial choropleth map (S1, S2), and grouped dotplot (G1, G2). Two levels of uncertainty (low, high) are used with each example. Ignoring the points in R1 and R2, there are no differences between the high and low uncertainty versions. Ignoring uncertainty can lead to misrepresentation of data."

  reg1a + choro1 + uni1a + reg1b + choro2 + uni1b + plot_layout(ncol=3, widths=c(30, 40, 30))

  ~~~

- These examples will serve as touchstones for a discussion of uncertainty visualisation, focusing on the approaches suggested in the literature.
-
- ### Uncertainty as a statistic
- Uncertainty is also often treated as another statistic, for example, the exceedance probability map of spatial data (see @Kuhnert2018 with software available in @Lucchesi2021), where the probability of exceeding a specified value is displayed on the map, placing the focus on extreme events.
- For risk communication, @Spiegelhalter2017 recommends showing probability with a set of coloured icons to communicate the relative number of affected individuals. The summary plot [@Potter2010] was also developed to address concerns about uncertainty in data displays. This dense display combines the plot of a single set of numeric values with a density estimate, boxplot, and statistical moments.
+ These examples will serve as touchstones for a discussion of uncertainty visualisation, focusing on the approaches suggested in the literature.
+
+ ## Uncertainty as a statistic
+ To avoid ignoring uncertainty entirely, some authors opt to visualise uncertainty using an "uncertainty statistic".
+ Rather than visualise the statistic we wish to draw inference on, we calculate some statistic related to the uncertainty (such as a probability, variance, or error) and visualise that instead.
+ For example, in geospatial visualisation, we might see an exceedance probability map [@Kuhnert2018], where areas are coloured according to the probability that the estimate exceeds a specified value.
+ In risk communication, we might see an icon array (or pictograph) [@Spiegelhalter2017] that communicates the relative number of affected individuals for a particular treatment using coloured human icons.
+ In statistics, we have the summary plot of @Potter2010, which combines the plot of a single set of numeric values with a density estimate, boxplot, and statistical moments.
+
+ Authors take these approaches because they explicitly see uncertainty as a variable of importance in and of itself [@Blenkinsop2000].
+ This is a perfectly acceptable motivation as uncertainty is often a variable of importance, but should we consider these visualisations "uncertainty visualisations"?
+ Is making an uncertainty visualisation that easy?
+
+ ### Example: visualising variance
+ @fig-statistic depicts the six plots introduced in @fig-ignore, but instead of visualising the central value of the distribution, we visualise an uncertainty statistic.
+ Plots R1 and R2 show the residual plot of our linear regression, which is a common approach for isolating and evaluating the noise in a regression.
+ By changing the underlying statistic, we have completely changed the interpretation of our plot, and therefore, the plot cannot be evaluated using our signal modulation criteria.
+ While the divergence from the signal modulation goal is clear in the case of regression residuals, this is not always the case.
+ Plots S1 and S2 show an exceed probability map, where each county is coloured according to $\widehat{P}(temperature>27)$.
+ At first glance, we might think this plot is performing signal modulation as the sine wave trend is clearly visible when the error is low, but barely visible with high error; however, just like R1 and R2, the meaning of these plots was changed when we swapped the underlying statistic.
+ As a temperature of 27 is a middling value in our example, as the variance increases, $\widehat{P}(temperature>27)$ approaches 0.5 for all counties.
+ Exceedance probability maps are often used to identify extreme values, and if we replaced 27 with a more extreme value (such as 35), the signal would swap, and we would see the wave pattern in the high uncertainty case, but not the low uncertainty case.
+ Continuing this trend, G1 and G2 visualise the variance of the distributions, and once again show that changing the underlying statistic changes the fundamental meaning of our graphics, but with a slight twist.
+ This plot reveals some interesting information about our data: the variance is constant in G1 but different for each group in G2.
+ Here, we highlight what is *actually* going on when we visualise uncertainty as a statistic.
+ The approach allows us to glean new information about the variance itself, but this information is largely independent of the signal conveyed by @fig-ignore.

- Some authors take this approach because they explicitly believe uncertainty is a variable of importance [@Blenkinsop2000], while others straddle the line, asserting uncertainty is acting as signal and noise, and should fulfil both roles [@geointerviews]. Simply put, these approaches might be described as swapping out the statistic for an "uncertainty statistic" to get an "uncertainty visualisation". Is it really that easy?
-
- #### Example: visualising variance
- @fig-statistic depicts the six plots showing this approach for the data introduced in @fig-ignore. The original central value estimate has been replaced with an uncertainty statistic.
- Plots R1 and R2 show the residual plot of our linear regression instead of the scatterplot and the regression line.
- The visual patterns we are looking for in this plot are distinct from the linear regression, so it is hard evaluate it relative to the trend.
- Sometimes the variance approach is still related to our original display, as we can see in the exceedance probability maps depicted in plots S1 and S2. These map $P(temperature>27)$ to colour for each county. The sine wave trend is clearly visible when the error is low, but barely visible with high error.
- Our visualisation of the univariate groups does reveal something interesting: the variance is constant in G1 but different for each group in G2. It is a nonsensical display, though, because the trend has completely disappeared.
- In practice, two plots are typically presented: one showing the main estimate and the other showing the uncertainty.

  ```{r}
  #| echo: false
@@ -771,57 +791,87 @@ In practice, two plots are typically presented: one showing the main estimate an
  #| fig-height: 6
  #| fig-width: 10
  #| out-width: 100%
- #| fig-cap: "Treating the uncertainty as a statistic,  with the same six examples. The regression (R1 and R2) is a scatterplot of residuals vs explanatory variable, separated from the regression making comparison of uncertainty related to the trend more difficult. For the choropleth (S1, S2): instead of temperature, the probability of exceeding 27$^o$C is shown. This has the effect of highlighting (sine wave) trend in the low error data, and de-emphasising it in the high error data. The plots G1 and G2 have replaced the treatment effect with the variance on our treatment effect. It is a bit nonsensical, but we learn something interesting that was not seen earlier: the variance in G2 is not uniform like that in G1."
+ #| fig-cap: "Treating the uncertainty as a statistic,  with the same six examples. The regression (R1 and R2) is a scatterplot of residuals vs explanatory variable, separated from the regression, making comparison of uncertainty related to the trend more difficult. For the choropleth (S1, S2): instead of temperature, the probability of exceeding 27$^o$C is shown. This has the effect of highlighting (sine wave) trend in the low error data, and de-emphasising it in the high error data. The plots G1 and G2 have replaced the treatment effect with the variance of our treatment effect. It is a bit nonsensical, but we learn something interesting that was not seen earlier: the variance in G2 is not constant like it is in G1."

  reg2a + exceed1 + uni2a + reg2b + exceed2 + uni2b + plot_layout(ncol=3, widths=c(30, 40, 30))
  ```

- #### What is an uncertainty visualisation, then?
+ ### What is an uncertainty visualisation, then?
  What an uncertainty visualisation is or is not is one of the most pervasive divides in the literature.
- For example, @Leland2005 mentions that popular graphics, such as pie charts and bar charts, omit uncertainty. @Wickham2011 suggests their product plot framework, for area plots like bar charts and also histograms, needs to be extended to include uncertainty representation.
- However, pie charts, bar charts and histograms have all been used in a significant number of experiments as examples of an "uncertainty visualisation" [@Ibrekk1987; @Olston2002; @Zhao2023; @Hofmann2012].
- What is going on here?
-
- This conflict stems from a subconscious disagreement about the purpose of uncertainty visualisation.
- If you believe uncertainty visualisation is about communicating risks or random variables, uncertainty visualisations are just visualisations of "uncertainty statistics" or distributions.
- On the other hand, if you believe uncertainty visualisation is about suppressing false signals visually, then you see an uncertainty visualisation as a transformation of an existing graphic that adds the uncertainty in.
- The former has no limitation on the visual appearance of an "uncertainty visualisation", allowing pie charts, bar charts or histograms, so long as the graphic is visualising "uncertainty", while the latter believes uncertainty visualisations only exist in relation to some "normal" visualisation.
- When we refer to the graphics depicted in @fig-statistic as "uncertainty visualisations", we are classifying visualisations by the data they display, not their visual features.
- This is not the standard approach in statistical graphics.
+ For example, @Leland2005 mentions that popular graphics, such as pie charts and bar charts, omit uncertainty.
+ @Wickham2011 suggests their product plot framework, for area plots like bar charts and also histograms, needs to be extended to include uncertainty representation.
+ However, pie charts, bar charts, and histograms have all been used in a significant number of experiments as examples of an "uncertainty visualisation" [@Ibrekk1987; @Olston2002; @Zhao2023; @Hofmann2012].
+ These examples imply that a chart being classified as an "uncertainty visualisation" is entirely at the discretion of the author, which brings into question whether the category is meaningful at all.
+ How did such a conflict arise in the literature?
+
+ This conflict stems from an unrecognised disagreement about the purpose of uncertainty visualisation.
+ If you think uncertainty visualisation is about communicating risks or probabilities, uncertainty visualisations are just visualisations of "uncertainty statistics".
+ On the other hand, if you think uncertainty visualisation is about signal modulation, then you see an uncertainty visualisation as a transformation of an existing graphic that adds the uncertainty in.
+ The former has no limitation on the visual appearance of an "uncertainty visualisation", allowing pie charts, bar charts, or histograms, so long as the graphic is visualising "uncertainty", while the latter believes uncertainty visualisations only exist in relation to some "normal" visualisation.
+
+ While arguments have been made for both approaches, there are two key reasons we shouldn't think of visualisations of "uncertainty statistics" as uncertainty visualisations.
+ The first problem is that it goes against established norms in the field of visualisation.
+ When we refer to the graphics depicted in @fig-statistic as "uncertainty visualisations", we are classifying visualisations by the data they display, not by their visual features.
  A scatter plot that compares means and a scatter plot that compares variances are both scatter plots.
+ There is no reason for "uncertainty statistics" to deviate from these conventions.
+ The second problem is slightly more subtle, and that is that the "uncertainty" in these graphics is not acting as uncertainty at all; it is acting as signal.

- Unlike plots, which are not defined by a statistic, uncertainty is *only* defined in relation to our visual statistic, a topic that frequently appears in the literature to be dependent on the "goals" of our analysis.
- @Meng2014 commented that what is kept as data and what is tossed away is determined by the motivation of an analysis – what was previously noise can become signal depending on the question. @Otsuka2023 suggested that the process of observing data to calculate statistics is largely dependent on our goals, because the process of boiling real-world entities down into probabilities depends on the relationships we seek to identify within our data.
+ How do we know that the uncertainty in these visualisations is acting as signal and not as noise?
+ The role of signal and noise is not set in stone and can change depending on the goals of our analysis.
+ @Meng2014 commented that what is kept as data and what is tossed away is determined by the motivation of an analysis – what was previously noise can become signal depending on the question.
+ @Otsuka2023 argued that the entire process of boiling real-world entities down into data requires us to have an internal model, which he refers to as "probabilistic kinds", that are defined by the relationships we seek to identify.
  @Wallsten1997 argue that the best method for evaluating or combining subjective probabilities depends on the uncertainty the decision-maker wants to represent, and why it matters.
  @Fischhoff2014 suggested we should have methods for communicating uncertainty depending on what the user is supposed to do with it. @Spiegelhalter2017 says we "cannot assess the quality of risk communication unless the objectives are clear".
  @geointerviews asserted that whether or not uncertainty is a source of doubt depends on the context.
- The sentiment behind this repeated point is clear: the role of uncertainty or signal is not dependent on the "type" of statistic, on the source of the information, or the methods we use; it is determined by the statistic we wish to draw inference on.
- Therefore, the fundamental problem with the "uncertainty statistic" approach is that the uncertainty in the plot isn't acting as noise; it is acting as signal.
+ The sentiment behind these discussions is clear: the role of signal and noise is determined by the hidden parameter we are trying to identify; whether that parameter is a measure of central tendency or spread is largely irrelevant.
+ When we swap a linear regression out for a plot of our residuals, the goal of the visualisation, and therefore the roles of signal and noise, change as well.

- If the uncertainty in a graphic is acting as a signal, there isn't an interesting perceptual challenge associated with the visualisation: the uncertainty can be displayed using standard principles of graphic design.
- In changing the inferential statistic, we also haven't dealt with the original problem of integrating noise, as these "uncertainty statistics" *also have associated uncertainty in the estimates* (e.g. variance of standard deviation estimate) that is being ignored.
+ If the uncertainty in a graphic is acting as a signal, there isn’t an interesting perceptual challenge associated with the visualisation: the uncertainty can be displayed using standard principles of graphic design.
+ In changing the inferential statistic, we also haven't dealt with the original problem of integrating noise, as these "uncertainty statistics" *also have associated uncertainty in the estimates* (e.g., variance of standard deviation estimate) that is being ignored.
  There is nothing wrong with explicitly visualising variance, error, bias, or any other statistic.
  These metrics provide important and useful information for analysis and decisions.
  The problem with this approach is that it means everything is an uncertainty visualisation, and if everything is an uncertainty visualisation, nothing is.

- ### Uncertainty as a variable
- Another common characterisation of uncertainty is as just another variable to be integrated into the visualisation, which means uncertainty visualisation is, at its core, a high-dimensional visualisation problem [@Griethe2006; @geointerviews].
- This occurs within computer science [@Kinkeldey2014], cartography [@MacEachren2005], and statistical graphics [@Leland2005].
-
- Discussion of the visualisations focuses on how "integrated" the uncertainty is with the estimate.
- @Kinkeldey2014 identified a split between intrinsic plots, where we map uncertainty to the colour or size of the geometric object of our estimate, and extrinsic plots, where uncertainty is mapped to a separate geometric object, such as glyphs or error bars.
- Similarly, @uncertchap2022 classified uncertainty visualisations as graphical annotations (extrinsic), and probability mapped to a visual encoding channel (intrinsic), or a hybrid of the two.
- It is unclear if these levels of "integration" in a plot design affect its ability to suppress signals.
-
- #### Example: mapping two independent variables
- @fig-variable shows the six examples introduced in @fig-ignore. Here, uncertainty is mapped to a spare aesthetic in the plot. In the grammar of graphics, variables are mapped to aesthetics, like position, colour, and size, within a plot.
- In plots R1 and R2, the standard error of the slope is represented by the width of the line. This is common, but it fails to represent the standard error of the intercept alongside the slope. We might think we can examine the width of the line at `wt=0`, but this would be incorrect. Here, it has been included, less obviously, by mapping the standard error of the intercept to transparency. While these help to give a gestalt of the uncertainty, they are not exacting
- representations. One would expect that the width above and below the line is one standard error, but why not map the width to two standard deviations, or even three? Interpreting transparency into a numerical quantity is also virtually impossible, so using this aesthetic mapping is effectively useless.
-
- A bivariate colour palette map is shown in plots S1 and S2. With two dimensions of the plot reserved for spatial position, it is difficult to incorporate the error. The bivariate colour palette maps the estimate to hue and the error to saturation, keeping the signal and noise contained to one visual aesthetic. This has the unfortunate effect of making the signal appear stronger when the error is higher (S2). Colour perception is a wild beast that is hard to tame.
-
- The extrinsic approach, shown in plots G1 and G2, has the estimate represented by a point, and uncertainty computed as a 95% confidence interval mapped to line length. The trend is still visible in both displays. It could be argued that where the variance is high, the trend remains a main focus; that is, the display fails to sufficiently suppress the signal.

+ ## Uncertainty as a variable
+ It is reasonable to assume that the "uncertainty as a statistic" approach falls short because we did not visualise the uncertainty within the context of our estimate.
+ This hypothesis leads to another common approach to uncertainty visualisation, where uncertainty is seen as an additional variable that needs to be integrated into the original visualisation of our estimates.
+ Under this philosophy, uncertainty visualisation becomes a question of how we can add additional information (uncertainty) into an existing visualisation without cluttering the plot and overwhelming the viewer [@Griethe2006; @geointerviews].
+ This approach emerges in computer science [@Kinkeldey2014], cartography [@MacEachren2005], and statistical graphics [@Leland2005] alike.
+ This "free aesthetic" approach to uncertainty visualisation encourages us to map any standard error to any unused aesthetic, with the primary concern being the intuitiveness of the mapping [@Maceachren2012].
+
+ When uncertainty is regarded as a variable, discussions focus on how integrated the uncertainty is with the estimate.
+ @Kinkeldey2014 identified a split between intrinsic plots, where we map uncertainty to colour or size of the geometric object of our estimate, and extrinsic plots, where uncertainty is mapped to a separate geometric object, such as glyphs or error bars.
+ Similarly, @uncertchap2022 classified uncertainty visualisations as graphical annotations (extrinsic), probability mapped to a visual encoding channel (intrinsic), or a hybrid of the two.
+ We can reframe these two approaches within the grammar of graphics.
+ In the grammar, *geometries* are the geometric objects we draw on the plot, while *aesthetics* represent things like position, colour, and size of said geometric objects.
+ In this framework, extrinsic approaches manage uncertainty at the geometry stage, while intrinsic approaches manage uncertainty at the aesthetic stage.
+
+ While the intrinsic versus extrinsic dichotomy is interesting, it is unclear if these levels of "integration" have any effect on the plot's ability to perform signal modulation.
+
+ ### Example: mapping two variables
+ @fig-variable shows the six examples introduced in @fig-ignore, with uncertainty mapped using an intrinsic or extrinsic approach.
+
+ In plots R1 and R2, we take an intrinsic approach to visualising uncertainty, where the estimated standard error of the slope is represented by the width of the line, while the error of the intercept is mapped to transparency.
+ While these help to give a gestalt of the uncertainty, they are not exacting representations.
+ Typically, we would visualise the uncertainty in a linear regression using a ribbon geometry, which would allow the uncertainty to change with respect to the wt variable.
+ By opting to use an intrinsic method, we highlight a key limitation of this approach: it assumes the uncertainty is constant for each geometric object.
+ Not only does this prevent the uncertainty from varying with x, but it also prevents us from displaying multiple standard errors.
+ Here we have used one standard error, but why not map the width to two standard errors, or even three?
+ The obfuscation of the variance in the slope is not the only issue with this intrinsic mapping.
+ We mapped the standard error of the intercept to transparency, as it is a highly intuitive mapping; however, translating this transparency to a numerical quantity is virtually impossible.
+ While this approach renders this aesthetic mapping effectively useless, this is the mapping that this design approach suggests.
+
+ Another intrinsic approach is illustrated in the bivariate colour palette maps shown in plots S1 and S2.
+ With two dimensions of the plot reserved for spatial position, it is difficult to incorporate the error.
+ The bivariate colour palette maps the estimate to hue and the error to the saturation and brightness, keeping the signal and noise contained to one visual aesthetic.
+ In this case, even though the signal and noise are mapped within a single visual channel (colour), the wave pattern remains visible in both plots.
+
+ The extrinsic approach, shown in plots G1 and G2, has the estimate represented by a point, and uncertainty computed as a 95% confidence interval mapped to line length.
+ Our plot now clearly shows the change in variance in the high uncertainty case, but the incremental trend in the average is still visible.
+ Therefore, the display fails to sufficiently suppress the high uncertainty signal.
+
+ We can see that all three approaches are unsuccessful when we evaluate their ability to perform signal modulation.
  Because all of these graphics visualise the distribution's estimate and uncertainty as two separate pieces of information, the message of the plot is "here is the trend *and* here is the uncertainty".
  It is worthwhile to examine why this occurs, to see if we can move towards a version of this plot where we are able to communicate signal and noise simultaneously.

@@ -833,31 +883,33 @@ It is worthwhile to examine why this occurs, to see if we can move towards a ver
  #| fig-height: 6
  #| fig-width: 10
  #| out-width: 100%
- #| fig-cap: "Treating the uncertainty as a variable, with the same six examples. In plots R1 and R2, the standard error of the slope is mapped to the line width. The standard error of the intercept is mapped to the transparency, which is less conventional. In plots S1 and S2, a bivariate colour palette is used with mean mapped to the hue, and error mapped to saturation. Plots G1 and G2 represent the mean of each group as a point, and the variance as an interval. The uncertainty is integrated well with the signal, but for the choropleth, the result is undesirable: the signal is easier to see when the error is higher."
+ #| fig-cap: "Treating the uncertainty as a variable, with the same six examples. In plots R1 and R2, the standard error of the slope is mapped to the line width. The standard error of the intercept is mapped to the transparency, which is less conventional. In plots S1 and S2, a bivariate colour palette is used with mean mapped to the hue, and estimated standard error mapped to saturation and brightness. In plots G1 and G2, the mean of each group is visualised as a point, and the standard error is visualised as an interval. In this example, the uncertainty has been integrated with the signal, but the results are undesirable. This is especially true in the case of the choropleth map, where the signal is easier to see when the error is higher."

  reg3a + bivar1 + uni3a + reg3b + bivar2 + uni3b + plot_layout(ncol=3, widths=c(30, 40, 30))

- #### Can we visualise a "single integrated uncertain value"?
+ ### Can we visualise a "single integrated uncertain value"?
  The reality is, based on our discussion on inferential statistics, uncertainty *isn't* a separate variable: it is a component of the random variable that is indistinguishable from the random variable itself.
- Similarities between the "as a variable" approach and the "as a statistic" approach are apparent when we read motivations for visualising an estimate (i.e. @fig-ignore) and variance (i.e. @fig-statistic) side-by-side using two separate graphics. This organisation is vulnerable to change blindness [@simons1997] as one needs to switch focus between two displays.)
- The preference for the methods utilised in @fig-variable is usually motivated by the difficulties in combining information on two separate graphics [@moritz2017trust; @Correll2018], rather than an understanding that the "uncertainty statistic" approach is not philosophically sound.
- To achieve signal suppression, we need to visualise noise and signal together as a “single integrated uncertain value” [@Kinkeldey2014] rather than as two separate statistics.
+ Similarities between the "as a variable" approach and the "as a statistic" approach become apparent when investigating why some authors opt for the variable approach in the first place.
+ When visualising a point estimate (i.e., @fig-ignore) and variance (i.e., @fig-statistic) side-by-side, one needs to switch focus between the two displays, which makes it difficult to combine the information on the graphics [@moritz2017trust; @Correll2018].
+ This implies that the reason authors stop visualising uncertainty as a statistic and instead visualise uncertainty as a variable is because of the cognitive burden imposed by switching focus, rather than the understanding that the method is not philosophically sound.
+ To achieve signal modulation, we need to visualise noise and signal together as a “single integrated uncertain value” [@Kinkeldey2014] rather than as two separate statistics.

  Just because we can still see the signal in @fig-variable, that does not mean the reading of the estimate is completely independent of the uncertainty.
- When making any visualisations, we usually want the visual channels to be separable, that is, we don't want the data represented through one visual channel to interfere with the others [@Smart2019]. It is also interesting to know whether readers can see all structures when there are multiple structures present in a data plot, or whether they systematically fixate on one [@vanderplas2017].
+ When making any visualisation, we usually want the visual channels to be separable, that is, we don't want the data represented through one visual channel to interfere with the others [@Smart2019].
  Separability may be desirable in standard data visualisation, but in uncertainty visualisation, it allows the estimate and its variance to be read independently, potentially leading to the uncertainty being ignored [@uncertchap2022].
  Therefore, rather than trying to maintain visual separability, the goals of uncertainty visualisation align far better with the pursuit of visual integration.
  In an ideal system, our estimate and uncertainty would be manipulated separately, but would be so *well-integrated that they are read as a single channel by the human brain*.
- The problem is that even if we can implement the most extreme versions of integrable, our methods fall short, as illustrated by the bivariate colour palette map in @fig-variable.
- Colour hue and brightness are one of the classic examples of integrable variables [@Vanderplas2020], and decreasing saturation should make the colours harder to distinguish, but the signal is still clearly visible with the high variance.
- This is to say nothing of the fact that multi-dimensional colour palettes can make the graphics harder to read and less accessible [@Vanderplas2015].
+ The problem is that even when mapping uncertainty and estimate to the same aesthetic, as illustrated by the bivariate colour palette map in @fig-variable, our methods fall short.
+ Colour hue and brightness are classic examples of integrable variables [@Vanderplas2020], and decreasing saturation should make the colours harder to distinguish, but the signal is still clearly visible when the variance is large.
+ Worse, multi-dimensional colour palettes can make the graphics harder to read and less accessible [@Vanderplas2015].

- #### Another example: mapping combined variables
- The Value Suppressing Uncertainty Palette (VSUP) [@Correll2018] was designed with the intention of preventing high uncertainty values from being extracted from a map by blending colours together as they become less certain.
- @fig-vsup shows the spatial example (S1, S2) using the VSUP approach.
- Since the palette was designed with the extraction of individual values in mind and it has only been tested on simple value extraction tasks [@Correll2018] or search tasks [@Ndlovu2023], we can see that, at least for our example, when the uncertainty is high, the spatial trend has functionally disappeared.
+ ### Mapping combined variables
+ The Value Suppressing Uncertainty Palette (VSUP) [@Correll2018] was designed with the intention of preventing high uncertainty values from being extracted from a map, and this is achieved by blending colours together as they become less certain.
+ @fig-vsup shows the spatial example (S1, S2) using this approach.
+ Since the palette was designed with the extraction of individual values in mind, it has only been tested on simple value extraction tasks [@Correll2018] or search tasks [@Ndlovu2023].
+ We can see that, at least for our example, when the uncertainty is high, the spatial trend has functionally disappeared.

  ```{r}
  #| echo: false
@@ -867,7 +919,7 @@ Since the palette was designed with the extraction of individual values in mind
  #| fig-height: 4.5
  #| fig-width: 9
  #| out-width: 80%
- #| fig-cap: "The spatial examples displayed with a choropleth map using a VSUP colour palette, where hue is blended when increased uncertainty. Plot B has successfully produced signal suppression. Look closely at the scales, though: it may have another explanation."
+ #| fig-cap: "The spatial examples displayed with a choropleth map using a VSUP colour palette, where hue is blended when uncertainty is increased. Plot B has successfully produced signal suppression. Look closely at the scales, though: it may have another explanation."

  vsup1 | vsup2

@@ -876,19 +928,22 @@ vsup1 | vsup2
  Finally, we have signal suppression!
  Well, not really, sorry, we tricked you.
  The two plots depicted in @fig-vsup actually show the exact same data; they are both the low variance case.
- If you look closely, you can see that the two plots have a different scale, where plot A has been scaled according to our existing knowledge about this data, while plot B has been scaled using the range of the data passed to the plot.
+ If you look closely, you can see that the two plots have a different scale.
+ Plot A has been scaled manually using the maximum and minimum error from the high and low uncertainty cases, while plot B has been scaled automatically using only the data from the low uncertainty case, which is the only data that would be visible in the plot.
  Since the variance and estimate are scaled independently, arbitrary differences in the range of our variance, unrelated to the estimate itself, will have significant impacts on the visual appearance of our plot.
  The scale issue in VSUP maps was also recognised by @Kay2019, who noted that the suppression of any one hypothesis largely depends on the methods we use to combine the palette, and the variance levels at which the blending occurs.
- This means that, for us to know that our plot will successfully perform signal suppression, we need to already know what signal we are trying to suppress and set up the VSUP palette accordingly.
+ This means that, for us to know that our plot will successfully perform signal validation, we need to already know what signal we are trying to validate and set up the VSUP palette accordingly.
  This means that VSUP maps are not suitable for exploratory data analysis.

- #### Uncertainty and exploratory data analysis
+ ### Uncertainty and exploratory data analysis
  The lack of uncertainty in descriptive statistics is due to the lack of inference.
- Descriptive statistics are actually a small piece of a much larger field, exploratory data analysis (EDA), that tends not to perform statistical inference.
- @Tukey1977 described EDA as the process of searching for interesting hypotheses ("the greatest value of a picture is when it forces us to notice what we never expected to see"), and defined it in relation to confirmatory data analysis (CDA), the process of verifying a hypothesis. There are more subfields of EDA: initial data analysis [@huebner2016; @chatfield], which involves checking assumptions and data quality prior to CDA, and model diagnostics (e.g. @bkw1980), including posterior checks of model fit. What binds these pursuits together is their reliance on visual summaries for making assessments and an absence of formal inference.
+ Descriptive statistics are actually a small piece of a much larger field, exploratory data analysis (EDA), that tends to de-emphasise statistical inference.
+ @Tukey1977 described EDA as the process of searching for interesting hypotheses ("the greatest value of a picture is when it forces us to notice what we never expected to see"), and defined it in relation to confirmatory data analysis (CDA), the process of verifying a hypothesis.
+ Subfields of EDA include: initial data analysis [@huebner2016; @chatfield], which involves checking assumptions and data quality prior to CDA, and model diagnostics (e.g., @bkw1980), including posterior checks of model fit.
+ What binds these pursuits together is their reliance on visual summaries for making assessments and an absence of formal inference.

-  @Hullman2021 argued that the EDA and CDA are not entirely distinct, as it is often difficult to draw a hard line.
- Our belief, as with many concepts, is that these approaches exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached.
+ @Hullman2021 argued that the EDA and CDA are not entirely distinct, as it is often difficult to draw a hard line.
+ Our belief is that these exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached.
  It can help to think of the knowledge-generating process of EDA and CDA as the nozzle on a hose with multiple spray options, where EDA is a fine misting spray that touches everything in the room, and CDA is a high-pressure jet capable of obliterating any and all debris from any single spot.

  Viewing EDA and CDA as a dichotomy can create some confusion when it comes to understanding the source of uncertainty in our analysis.
@@ -898,30 +953,42 @@ One of the trade-offs in the two methods is that deductive conclusions provide c
  This means that EDA, the inductive counterpart, has uncertainty in any conclusions reached, with the requirement to follow up our newfound hypothesis with CDA if we want true certainty.
  This adds another layer of confusion to the study of uncertainty visualisation: conflating the uncertainty in our data with the uncertainty that is inherent to an exploratory process (EDA).

- Authors often over-compensate for the inherent EDA uncertainty, pre-emptively hedging against every false inference that could possibly be drawn from a graphic. @Hullman2021 argues there is no such thing as a "model-free" visualisation. @Guo2024 provides a grammar for visualising statistical model checks. The lineup protocol [@Buja2009] provides the viewer with plots of the data in a field of plots of null data where any patterns seen are due to sampling variability. The Rorschach protocol, from the same paper, shows only null plots to give the reader some intuition for what spurious sampling patterns exist. @Savvides2019 provides a statistical super-test against multiple comparisons driven by probabilistic arguments. There is a CDA quality to these approaches. The sum total of a lot of CDA is not EDA, just as swinging a high-pressure jet around a room is not equivalent to using a misting spray. While EDA and CDA may be along a continuum, we cannot simultaneously perform EDA and CDA as the approaches are, philosophically speaking, perpendicular to one another.
+ Authors often over-compensate for the inherent EDA uncertainty, pre-emptively hedging against every false inference that could possibly be drawn from a graphic.
+ @Hullman2021 argues there is no such thing as a "model-free" visualisation. @Guo2024 provides a grammar for visualising statistical model checks.
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+ The Rorschach protocol, from the same paper, shows only null plots to give the reader some intuition for what spurious sampling patterns exist.
+ @Savvides2019 provides a statistical super-test against multiple comparisons driven by probabilistic arguments. There is a CDA quality to these approaches.
+ The sum total of a lot of CDA is not EDA, just as swinging a high-pressure jet around a room is not equivalent to using a misting spray.
+ While EDA and CDA may be along a continuum, this comes from the large number of hypotheses that can be identified using visualisation.
+ For any particular hypothesis, we cannot simultaneously perform EDA and CDA, as doing so would require us to violate causality.
+ It is impossible to validate a hypothesis before we even know it exists.

  To truly create an uncertainty visualisation approach that is capable of EDA, we need to accept that uncertainty is inherent to the method, and it cannot be pre-emptively removed from the visualisations.
  If we accept this fact, then the only possible source of uncertainty in an uncertainty visualisation system that performs EDA is from the data itself.
- That is to say, the only possible way for there to be uncertainty in a visualisation designed for EDA is that the data itself represents inference that has been done earlier in our analysis.
+ That is to say, the only possible way for there to be uncertainty in a visualisation designed for EDA is that the data itself is a summary resulting from inference that has been done earlier in our analysis.
  We can see this in the original description of @fig-ignore, where the uncertainty in all cases, even measurement error, represents inference that was performed earlier in our analysis.

- In the VSUP approach, our ability to arbitrarily decide which values to blend at, or which suppression approach to use, means that the "uncertainty" we are visualising will be informed by the conclusions we are drawing, and not a product of the data itself -- an antithetical approach to EDA.
+ In the VSUP approach, our ability to arbitrarily decide which values to blend at, or which suppression approach to use, means that the "uncertainty" we are visualising will be informed by the conclusions we are drawing, and not a product of the data itself.
+ This is an antithetical approach to EDA.
  For a visualisation to be suitable for EDA, it should always look the same regardless of what hypothesis we plan to draw from it.
  As much of the transparency in data visualisation comes from this feature in EDA, it is reasonable to set it as a requirement of our visualisations.
- Ensuring this property means we cannot treat signal and noise as separate variables, but rather as a single integrated unit.
+ This means we cannot treat signal and noise as separate variables, but rather we must treat them
```

~~~diff
+ This means we cannot treat signal and noise as separate variables, but rather we must treat them as a single integrated unit.

- ### Uncertainty as a distribution
+ ## Uncertainty as a distribution
  Rather than trying to reduce random variables to a single value or pair of values, why not visualise the whole distribution?
- This approach is found in computer science's hypothetical outcome plots (HOPs), which animate a sequence of potential outcomes of a distribution [@Hullman2015], in geoscience's pixel-maps [@Lucchesi2021; @Blenkinsop2000], and in statistics multiple forecasts [@fpp3].
-
- #### Example: visualising samples
- @fig-variable shows the six examples introduced in @fig-ignore, shown as distributions. The distribution is represented by either a sample of outcomes or a quantile dot plot.
- Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. The distribution used for both the slope and intercept is normal with the conventional mean and standard error.
- Plots S1 and S2 show a pixel map, which is a choropleth map where each county is coloured by a sample of outcomes from the distribution.
+ This approach is found in hypothetical outcome plots (HOPs), which animate a sequence of potential outcomes of a distribution [@Hullman2015], in pixel-maps [@Lucchesi2021; @Blenkinsop2000], and in visualisations of multiple forecasts [@fpp3].
+
+ ### Example: visualising samples
+ @fig-distribution shows the data from the six examples introduced in @fig-ignore, as distributions.
+ The distribution is represented by either a sample of outcomes or a quantile dot plot.
+ Plots R1 and R2 show a linear regression as a sample of possible outcomes from the estimated bivariate normal distributions of the slope and intercept.
+ Plots S1 and S2 show a pixel map, which is a choropleth map where each county is subdivided into an $n\times n$ grid.
+ Each grid cell is coloured according to a simulated outcome from the county's distribution, giving the map a pixelated appearance.
  Plots G1 and G2 show each univariate distribution as a quantile dot plot.
+ Quantile dotplots plot a uniform sample of quantiles as a dotplot, offering a discrete alternative to density plots [@Kay2023].
  We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the univariate distributions are all clearly visible in the low variance case, but disappear in the high variance cases.
- The graphics have achieved signal suppression.
- Visualising the random variable as a distribution gives additional information, such as the previously hidden bimodality of the univariate distributions in G1 and G2.
+ The graphics have achieved signal validation.
+ Visualising the estimate as a distribution also gives additional information, such as the previously hidden end peaks of the univariate distributions in G1 and G2.


  ```{r}
@@ -932,62 +999,66 @@ Visualising the random variable as a distribution gives additional information,
  #| fig-height: 6
  #| fig-width: 10
  #| out-width: 100%
- #| fig-cap: "Treating the uncertainty as a distribution, with the same six examples. Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. Plots S1 and S2 show a pixel map. Plots G1 and G2 show the groups as quantile dot plots. In each case, the signal (regression line, sine wave, increasing trend) has disappeared with high uncertainty."
+ #| fig-cap: "Visualising the uncertainty as draws from an estimated distribution, with the same six examples. Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. Plots S1 and S2 show a pixel map. Plots G1 and G2 show the groups as quantile dot plots. In each case, the signal (regression line, sine wave, increasing trend) has disappeared with high uncertainty."

  reg4a + pixel1 + uni4a + reg4b + pixel2 + uni4b + plot_layout(ncol=3, widths=c(30, 40, 30))


  ~~~

- #### Quantified versus unquantified uncertainty
- By showing our distribution as "data", we are able to read the "uncertainty" plots using the same perceptual mechanisms we use to read the "non-uncertainty" plot.
+ ### Quantified versus unquantified uncertainty
+ By showing our distribution as "data", we are able to read the "uncertainty" plots (i.e., @fig-distribution) using the same perceptual mechanisms we use to read the "non-uncertainty" plots (i.e., @fig-ignore).
  This should lead to more effective communication, as people tend to read more complicated visualisations like the bivariate and VSUP plots the same way they read the simple choropleth counterpart [@Ndlovu2023].
  This approach also does not significantly hinder our ability to extract the individual values mapped by the previous plots, as extracting global statistics from a sample can be done with relative ease [@Franconeri2021].
- We are able to include more information by offloading more computation to visual processing, but what is the limitation on this approach?
+ For example, in @fig-distribution, even though we have not included the average in G1 or G2, a human viewer can still extract the average effect of each group with relative ease.
+ We are able to include more information by offloading more computation to visual processing, but what is the limitation of this approach?
  Which aspects of uncertainty should be processed by our visual system, and which should be processed by our statistical computation?
- It is not obvious from the question, but this is actually a question about how much of our uncertainty should be quantified.
+ This is actually a question about how much of our uncertainty should be quantified.

- Quantified uncertainty usually focuses narrowly on concepts such as probability, confidence intervals, variance, error, or precision [@Hullman2018; @Maceachren2012; @Thomson2005], while unquantified uncertainty often includes a broader range of concepts like missing values, reliability, model validity, or source integrity [@Griethe2006; @Leland2005; @Pang1997; @Pham2009; @Boukhelifa2017].
- When discussing uncertainty, we typically include these unquantified uncertainties, not because these things *are* uncertainty, but because they can *create* uncertainty when we perform inference.
- This is often because these unquantified uncertainties violate our assumptions of the uniformity of nature [@Otsuka2023].
+ Quantified uncertainty usually focuses narrowly on concepts that can be calculated from theoretical or empirical distributions, such as probability, confidence intervals, variance, error or precision [@Hullman2018; @Maceachren2012; @Thomson2005; @Kay2023].
+ On the other hand, unquantified uncertainty often includes a broader range of concepts like missing values, reliability, model validity, or source integrity [@Griethe2006; @Leland2005; @Pang1997; @Pham2009; @Boukhelifa2017].
+ When discussing uncertainty, we typically consider these unquantified uncertainties, not because these things *are* uncertainty, but because they can *create* uncertainty when we perform inference.
+ This is often because these unquantified uncertainties violate our model assumptions.
+ Quantifying the effect of these assumption violations on our estimates transforms unquantified uncertainty into quantified uncertainty.

  Sometimes we are able to visualise these assumption violations directly.
  For example, we can check for structure in our missing data using the `naniar` package @Tierney2023 that allows us to include missing values as a "shadow" alongside our usual visualisations.
  This approach amounts to just "showing the data", which is a simple but effective option for uncertainty visualisation that is largely overlooked.
- While this approach is useful for better understanding data, it will not eliminate trends that have become invalid due to structure in missing data or an invalid model.
+ While this approach is useful for better understanding data, it will not eliminate visual signals that are rendered invalid due to incorrect assumptions about the model or missing data.
  We can only integrate uncertainty as noise when that uncertainty has been *quantified* as an effect on the estimates we are visualising.
  This is not to say one method is preferable; visualising both quantified and unquantified uncertainty is necessary for a healthy analysis.
  Data analysis often works in cycles, where we find assumption violations using EDA, quantify the effect of these violations on inference, and then visualise the output of that inference using uncertainty visualisation.

- ## Evaluating uncertainty visualisations
- Unfortunately, the conflicting results in the field are not limited to plot design and extend to the experimental findings as well [@MacEachren2005; @Kinkeldey2014; @Hullman2016].
- There are as many explanations for the noisy evaluation studies as there are contradictions in the research itself.
- @Kim2019 believes there is some interference in results from participants' prior beliefs;
- @Hullman2016 believes the noise in the literature could come from visual heuristics, subjective probabilities, unknown participant utility functions, or a misunderstanding of statistical concepts (such as confidence intervals).
+ # Evaluating uncertainty visualisations
+ Unfortunately, the conflicting results in the field are not limited to plot design and extend to the experimental findings as well.
+ Reviews that attempt to synthesise evaluation studies identify conflicting results, such as: inconsistency in the best aesthetic mappings for uncertainty [@Kinkeldey2014; @MacEachren2005]; inconsistency across different contexts [@Kinkeldey2014; @Hullman2016]; and inconclusive evidence for whether uncertainty is helpful at all [@MacEachren2005].
+
+ There are as many explanations for these contradictions as there are contradictions themselves.
+ @Kim2019 suggests there is some interference in results from participants' prior beliefs;
+ @Hullman2016 notes that the irreconcilable evaluation studies are likely a product of visual heuristics, subjective probabilities, unknown participant utility functions, or a misunderstanding of statistical concepts (such as confidence intervals).
  @Kinkeldey2014 suggest the perception of visualisation changes by audience, so we cannot expect the same results between different subpopulations.
  @Brennen2018 attributes evaluation difficulties to cognitive load from complicated uncertainty visualisations, as well as the participants' prior experience in the topic.
- While these issues will certainly have some impact on our ability to synthesise, none of them is unique to uncertainty visualisation.
- Rather, the issue is likely due to a disconnect between the evaluation methods used and the stated goals of each experiment, a common issue in visualisation evaluation experiments [@Vanderplas2020].
+ While these issues will certainly have some impact on our ability to synthesise information, none of them is unique to uncertainty visualisation.
+ Rather, the issue is likely due to a disconnect between the evaluation methods used and the stated goals of each experiment, a common issue in visualisation evaluations [@Vanderplas2020].

- ### Current evaluation methods
+ ## Current evaluation methods

- #### Value extraction
+ ### Value extraction
  Uncertainty visualisations are most commonly evaluated based on how accurately viewers can extract an estimate and its variance [@Kinkeldey2014; @Hullman2019].
  This is not unusual, as direct observation is the simplest way to verify that information can be accurately read from a graph [@Vanderplas2020].
  Unfortunately, this approach doesn't work for uncertainty visualisation.
- The second we ask a specific question about a statistic, that statistic becomes inferential, even if the plot was not the intent behind the question.
- By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X}<x)$, we end up evaluating visualisations on their ability to convey uncertainty statistics, rather than their ability to perform signal suppression.
- Even if the authors do not realise it themselves, there is nothing unique to uncertainty in these studies, so when we boil the findings down to generalised results, they simply restate existing principles within information visualisation.
+ Once we ask a specific question about a statistic, that statistic becomes inferential, even if the plot was not designed for this purpose.
+ By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X}<x)$, we end up evaluating visualisations on their ability to convey uncertainty statistics, rather than their ability to perform signal modulation.
  Some of the findings are obvious: participants were more accurate when reading a probability expressed as text than when they had to extract it from a graphic [@Cheong2016; @Savelli2013].
  Other studies replicate existing research, such as the finding that a probability mapped to a position is more accurate than one mapped to an area [@Ibrekk1987; @Gschwandtnei2016], established as part of the hierarchy of perceptual tasks more than 40 years ago [@Cleveland1984; replicated by @heer2010].
- This extends beyond simple accuracy evaluations: @Sanyal2009 found that colour was more effective than size when searching for extrema in variances; we have known that pre-attentive aesthetics, such as colour, are more efficient for search tasks since the 1980s [@Vanderplas2020].
+ This extends beyond simple accuracy evaluations: @Sanyal2009 found that colour was more effective than size when searching for extrema in variances; we have known that pre-attentive (subconsciously processed) aesthetics, such as colour, are more efficient for search tasks since the 1980s [@Vanderplas2020].
  By classifying these studies as evaluations of "uncertainty visualisation" while evaluating uncertainty as a signal, we are encouraged to see successful examples of signal suppression as failure.
  This approach leads authors to advise against particular aesthetic mappings for uncertainty, because they cause participants to have more difficulty extracting values [@Blenkinsop2000].
- This conclusion is antithetical to the goals of signal suppression and occurs because these methods evaluate uncertainty as a signal, not as noise.
+ This conclusion is antithetical to the goals of signal validation and occurs because these methods evaluate uncertainty as a signal, not as noise.

- #### Trust, confidence, and risk aversion
- If we cannot directly measure uncertainty for fear that it turns into a signal, we might then assume we can measure the secondary benefits of increased transparency.
+ ### Trust, confidence, and risk aversion
+ If we cannot directly measure the effects of visual uncertainty for fear that it turns into a signal, we might instead try to measure the emotional or behavioural effect this uncertainty should have on the viewer.
  This seems to be the approach of many visualisation authors, as secondary benefits such as trust, confidence, and risk aversion are all frequently used in uncertainty evaluation studies [@Hullman2019].
  Unfortunately, measuring these secondary effects often leads to confusing conclusions that simultaneously argue for and against the inclusion of uncertainty.

@@ -997,10 +1068,11 @@ This sentiment is also true for confidence, as @Blenkinsop2000 commented that vi

  Another secondary effect that is similar to trust and confidence is risk aversion.
  Risk aversion is an economic term used to describe an agent who would choose a random variable with a lower expected payout because it also has a lower variance.
- Risk aversion's role in the uncertainty visualisation is unclear, as authors will argue uncertainty should elicit more risk aversion in one paper [@Hullman2019], and argue for less risk aversion (by proxy of suggesting rational agents as a benchmark) in the next [@wu2023rational].
- Ultimately, these approaches have similar issues to value extraction studies, except they are slightly more confusing in their goals, leading them to simultaneously argue for and against the inclusion of uncertainty in a visualisation.
+ There is some disagreement on the role of risk aversion in the literature, as @Hullman2019 argue uncertainty should elicit more risk aversion, whereas @wu2023rational argues for less risk aversion (by proxy of suggesting rational agents as a benchmark).
+
+ Ultimately, measuring these secondary benefits has similar issues to value extraction studies, except they are slightly more confusing in their goals, leading them to simultaneously argue for and against the inclusion of uncertainty in a visualisation.

- #### Alternative approaches
+ ### Alternative approaches
  Often, authors understand that the effects of uncertainty are more complicated than simple value extraction.
  These studies indicate that accurately capturing uncertainty will be more complicated than simply avoiding value extraction or trust as a measure.

@@ -1012,10 +1084,7 @@ On the other hand, questions that are incredibly specific about the distribution
  For example, @Hullman2015 asked participants to compare two normally distributed groups, A and B, and had many participants say that group A was more likely to be bigger, despite group B having a higher mean.
  @Gschwandtnei2016 asked participants the "probability that the interval has already ended at the marked point in time?" and participants replied with the probability that the interval had already started.

- The confusion around trying to capture the effects of uncertainty can also (understandably) extend to the authors of the study itself.
- We can see an example of this in @Padilla2017. In order to answer the question correctly, the first experiment required participants to assume that an oil rig being "more likely to be hit" by a hurricane would *not* translate to the rig sustaining "more damage". The second experiment required participants to assume the opposite.
-
- #### Effective methods
+ ### Effective methods
  This is not to say all evaluation studies fail to properly evaluate uncertainty as noise.
  There are several studies that ask participants to identify a particular signal that the noise is trying to obfuscate [@Kale2018; @Correll2014], which seems to be an effective method.
  The only problem with these studies is that most uncertainty visualisation methods are not integrated into *the grammar of graphics* [@Leland2005; @ggplot2], so we regularly see comparisons between disparate plots that would never be considered substitutes for one another outside the artificial "uncertainty visualisation" framework they are placed within.
@@ -1026,9 +1095,9 @@ This means that even when evaluations *are* done correctly, there is no generali
  The point here is not to accuse the authors of poor academic rigour.
  The papers are (usually) logically consistent and well-formulated pieces of work.
  Rather, the point is to illustrate that evaluating uncertainty as noise is surprisingly difficult.
- Designing tests for signal suppression will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.
+ Designing tests for signal validation will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.

- ### Implicit Hypothesis Testing
+ ## Implicit Hypothesis Testing
  The main problem with current uncertainty visualisation evaluations is that they often require explicit (or convoluted) questions about the variance.
  Asking direct questions about the statistics or outcomes is not an explicit requirement of visualisation evaluations.
  In their review of testing statistical graphics, @Vanderplas2020 drew a distinction between explicit tests, where participants are asked direct questions about specific features of a plot, and implicit testing, where users identify both the purpose and function of the plot.
@@ -1037,18 +1106,20 @@ Lineups are a confirmatory visualisation tool where participants are shown a set
  The implicit test does not limit the versatility of the approach, with the lineup being used to evaluate the effectiveness of different types of plots [@Hofmann2012], colour palettes [@Reda2021], and design decisions [@vanderplas2017].

  Lineup protocols are not only useful for implicit testing: they also have parallels to hypothesis testing that can be leveraged in uncertainty visualisation.
- The concept of signal suppression is, at its core, an assertion of statistical validity: the visibility of signals should be directly proportional to p-values or some equivalent measure.
- This comparison is not new in uncertainty visualisation, where parallels have been drawn to frequentist statistics by @Correll2014, who compared results to Cohen's D, and to Bayesian statistics by @Kim2019, who evaluated plots based on their impact on the users' prior beliefs.
- The comparison to hypothesis testing is far more natural for the lineup protocol, which has a visual test statistic [@Majumder2013] and can be compared to standard statistical tests using power curves [@Patrick2023].
- The connections between lineups and uncertainty visualisation are numerous and have been previously identified in the development of HOPs [@Hullman2015].
+ The concept of signal validation is, at its core, an assertion of statistical validity: the visibility of signals should be directly proportional to p-values or some equivalent measure.
+ This comparison is not new in uncertainty visualisation, where parallels have been drawn to frequentist statistics by @Correll2014, who compared results to Cohen's D, and to Bayesian statistics by @Kim2019, who evaluated plots based on their impact on the users' prior beliefs.
+ The comparison to hypothesis testing is far more natural for the lineup protocol, which has a visual test statistic and can be compared to results from Bayesian analysis [@vanderplas2017] or standard statistical tests using power curves [@Majumder2013; @Patrick2023].
+ The connections between lineups and uncertainty visualisation are numerous and have been previously identified in the development of Hypothetical Outcome Plots [@Hullman2015].

  The lineup protocol and uncertainty visualisations are similar: lineups were designed for checking if perceived patterns are real or merely the result of chance [@Buja2009; @Wickham2010].
  As both approaches are attempting to do the same thing, it is likely that we are unable to leverage the lineup protocol directly to evaluate uncertainty visualisation, but a new evaluation methodology should try to learn from the success of the lineup approach.
  Designing an implicit testing method for uncertainty visualisation that allows us to draw parallels to standard notions of statistical significance would solve many of the issues with the current evaluation approaches.

- ## Conclusions and Future Work
- This paper examines the literature and provides suggestions for a structural framework to support uncertainty visualisation. Particularly, we propose that uncertainty visualisation should accomplish signal suppression, dampening weak signals and amplifying strong signals. We have also highlighted several gaps in the existing literature.
+ # Conclusions and Future Work
+ This paper examines the literature and provides suggestions for a structural framework to support uncertainty visualisation.
+ Particularly, we propose that uncertainty visualisation should act as a signal modulation, suppressing weak signals, and amplifying strong signals.
+ Using this framework, there are four key gaps identified by this review.

  Experimental practices on uncertainty visualisation need standards.
  Some existing evaluation experiments treat uncertainty as a signal, while others treat uncertainty as noise.
@@ -1061,14 +1132,15 @@ When designing experiments, authors often choose aesthetics that are visually di

  Uncertainty needs to be formalised within the grammar of graphics.
  Some of this formalisation was done by @Kay2023, but it focuses only on the visualisation of univariate distributions.
- Giving authors the ability to describe uncertainty visualisations in terms of statistics, geometries and aesthetics will support evaluation experiments that can build towards a cohesive theory of visualising uncertainty.
+ Giving authors the ability to describe uncertainty visualisations in terms of statistics, geometries, and aesthetics will support evaluation experiments that can build towards a cohesive theory of visualising uncertainty.

- *Software that allows users to easily perform signal suppression is needed.*
+ *Software that allows users to easily perform signal validation is needed.*
  Existing uncertainty visualisation methods view a distribution as its own object, and there are no software options treating "an uncertainty visualisation as a function of an existing visualisation" philosophy.

- Signal suppression is an undeveloped area of visualisation research, and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation.
+ Signal modulation is an undeveloped area of visualisation research, and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation.
+

- ## Reproducibility {-}
+ # Reproducibility {-}

  The R packages used for this work were: `tidyverse` [@tidyverse], `RColorBrewer` [@RColorBrewer], `scales` [@scales], `sf` [@sf], `urbnmapr` [@urbnmapr], `flextable` [@flextable], `colorspace` [@colorspace], `ggdist` [@Kay2023], `ggdibbler` [@ggdibbler], `patchwork` [@patchwork], `distributional` [@distributional], `ggthemes` [@ggthemes], `broom` [@broom], and `rgeos` [@rgeos].
  The GitHub repository for this paper can be found at https://github.com/harriet-mason/ARSA-UncertaintyLitReview, which contains the files required to reproduce this article in full.
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